SALES PERFORMANCE ANALYSIS

Quick-Commerce Grocery Sales — Category, Segment, Region & Payment Mode Deep-Dive

Project Documentation

Dataset: zepto_sales_raw.xlsx | 1,500 orders | July -16, 2026

1. Problem Statement

Zepto (a quick-commerce grocery delivery platform) generates a continuous stream of order-level transaction data across cities, product categories, customer segments and payment modes. Business teams need a single, reliable view of how sales performance behaves across these cuts so that inventory, marketing and operations decisions can be made with evidence instead of guesswork.

Objective: Analyze the performance of products across four dimensions — category/sub-category, customer segment, region (city) and payment mode — and identify the underlying trend or seasonal pattern in sales, in order to recommend concrete actions that improve revenue and delivery reliability.

Specifically, the analysis was scoped to answer:

- Which product categories and individual SKUs drive the most revenue, and which underperform?
- Which customer segments (Office, Café, Cloud Kitchen, Household, Tea Stall, Bakery) contribute most to sales?
- How does performance vary by region (Bangalore, Delhi, Mumbai) and by payment mode (UPI, Cards, Wallets, COD, etc.)?
- Is there a visible month-on-month trend or seasonal pattern in order volume and revenue?
- What operational issues (cancellations, substitutions, refunds) are affecting realized revenue, and what should the business do about them?

2. Dataset Description

The source file is a raw transactional export, zepto_sales_raw.xlsx, containing 1,500 order-level records and 28 columns. Each row represents one order line item.

2.1 Structure

Field group	Columns	Description
Order & Customer	Order_ID, Order_Date, Customer_Name, Customer_Type, Customer_Segment	Order identifiers and who placed the order
Location	City, Pincode	Delivery region
Payment	Payment_Method	How the order was paid for
Product	SKU, Product_Name, Category	What was ordered (category acts as the category/sub-category cut)
Commercials	Qty, Price, Discount, Subtotal, Tax, Delivery_Fee, Surge_Fee, Total	Order value build-up
Fulfilment	Delivery_Slot, Warehouse, Rider_ID, Delivery_Status, Delivery_Time, Substituted	How the order was fulfilled
Post-order	Refund_Amount, Reason	Cancellations / returns and their cause

2.2 Coverage

- 1,500 orders, 300 distinct customers, 3 cities (Bangalore, Delhi, Mumbai) plus 28 orders with a missing city, grouped as N/A.
- Date range: 1 Jan 2025 – 1 Mar 2025 (~2 months of daily transactions; the March slice is a single day, not a full month).
- Reported (uncleaned) revenue across the Total column: ₹5,63,992; average order value ≈ ₹376; total tax collected ≈ ₹25,076 — closely matching the totals shown on the delivered dashboard (₹5,63,284.75 / ₹376.02 / ₹25,043.24), the small gap coming from currency-symbol and text formatting in a handful of raw cells that get normalized during cleaning.

3. Data Cleaning Process

The raw export is a deliberately messy real-world style extract. Before any aggregation or charting, the following data-quality issues were identified and corrected:

3.1 Issues found

Column	Issue	Example
Order_Date	3 different date formats mixed in the same column	2025-02-08, 15 Jan 2025, 21/01/2025
Payment_Method	Inconsistent casing/spacing; same method spelled differently	UPI, upi, UP I all mean UPI
Category	Mixed casing, typos, singular/plural variants	veg / Veg, frozen / Frozen, bevrage / Beverages, DIARY / Dairy, snaks / Snacks
Price	Currency symbol (₹) present in ~226 rows, absent in the rest, column stored as text	₹59.27 vs 94.25
City	28 blank values	Treated as region 'N/A' rather than dropped, to avoid losing order value
Unnamed column (between Category and Qty)	No header, inconsistent numeric codes not cleanly mapped to any other field	Retained only for traceability; excluded from analysis
Delivery_Time / Reason	Nulls for orders that were never delivered / never had an issue	Expected nulls — not treated as data errors

3.2 Cleaning steps applied

- Standardized Order_Date to a single ISO date type by detecting and parsing each of the three date patterns.

- Normalized Payment_Method by trimming whitespace and upper-casing, then collapsing look-alike spellings (UPI / upi / UP I → UPI).
- Normalized Category by trimming, lower-casing, and mapping typos/case variants to ten standard categories: Veg, Fruits, Dairy, Frozen, Snacks, Bakery, Beverages, Essentials, Baby, Other.
- Stripped the ₹ symbol from Price and cast the column to numeric.
- Kept blank City values as an explicit 'N/A' region bucket instead of deleting the rows, so revenue totals stay complete.
- Verified Order_ID uniqueness (no duplicate orders found) and left legitimately-empty fields (Delivery_Time, Reason, Refund_Amount) as nulls rather than imputing them, since a null there is meaningful (e.g. no refund reason because the order wasn't cancelled).
- Re-derived Subtotal/Tax/Total consistency as a spot check; no material formula errors were found once Price was numeric.

4. Dashboard Design

The cleaned dataset was used to build a single-page interactive dashboard so that category, segment, region and payment-mode performance can be reviewed at a glance, alongside the monthly trend.

4.1 Layout

- KPI strip: Total Revenue, Total Average (order value), Total Tax, and Total Number of Orders/Customers — the headline numbers for the period.
- Trends over Months (line chart): order value trend across Jan / Feb / Mar to expose the seasonal / time pattern.
- Sales of Each State (bar chart): regional performance across Bangalore, Delhi, Mumbai and N/A.
- City slicer (list): lets a viewer filter every visual by region in one click.
- Top 10 Product (bar chart): SKU-level performance — the sub-category cut requested in the brief.
- Top 10 Customer (bar chart): concentration of revenue among top accounts.
- Payment Method for Each Sales (pie chart): share of revenue by payment mode.

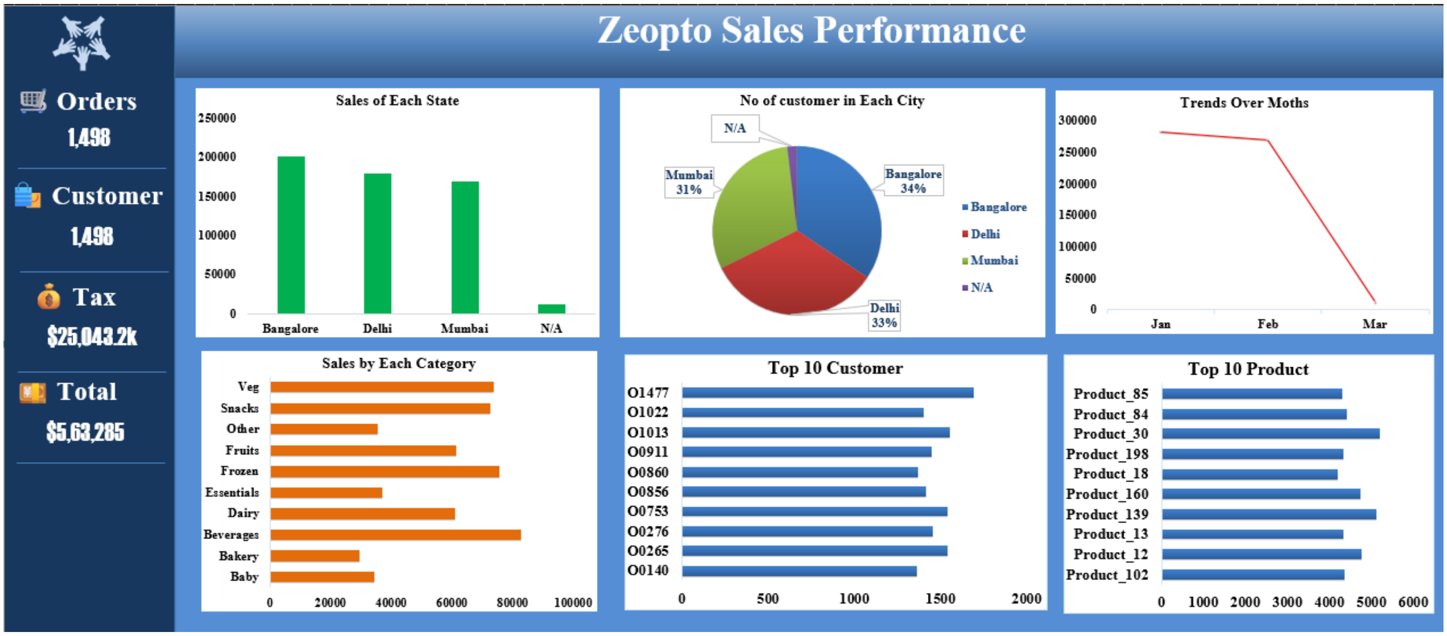

Figure 1: Delivered Sales Performance Analysis dashboard.

5. Key Insights

5.1 Category performance

Category	Revenue (₹)
Beverages	82,603
Frozen	76,044
Veg	73,582
Snacks	72,713
Fruits	61,713
Dairy	61,073
Essentials	36,991
Other	35,419
Baby	34,274
Bakery	29,579

- Beverages, Frozen and Veg are the three largest categories, together contributing roughly 40% of total revenue.
- Bakery and Baby are the smallest categories once typo-variants are merged into their correct bucket — before cleaning, their revenue was silently split across 2–3 spelling variants each, understating their true size.

- At the SKU level, a small set of products (Product_30, Product_139, Product_12, Product_160, Product_84…) each generate ₹4,300–5,200, well above the median product — a classic long-tail pattern where a handful of SKUs matter disproportionately.

5.2 Segment performance

- Office is the single largest customer segment by revenue (~₹1,03,700), followed closely by Café, Tea Stall and Household — the six segments are otherwise fairly balanced (₹88k–₹104k each), so no segment is negligible.
- By customer type, Bulk customers drive ~83% of total revenue (₹4,70,046 of ₹5,63,992) versus New and Returning customers combined (~17%) — the business is currently very dependent on a small number of high-volume 'Bulk' accounts.

5.3 Regional performance

- Bangalore leads (₹2,01,866), ahead of Delhi (₹1,80,571) and Mumbai (₹1,69,384); the gap between the top and bottom city is modest (~16%), so no city is dramatically under-served, but Bangalore is the growth benchmark for the other two.
- Orders with a missing city (~2% of orders, ₹12,171 in revenue) point to a data-capture gap at checkout or in a specific warehouse/app flow that should be fixed at the source.

5.4 Payment mode performance

- UPI is by far the dominant payment mode once its spelling variants (UPI / upi / UP I) are merged — together they represent ~33% of orders and ₹1,88,838 in revenue, more than double the next largest mode (NetBanking, ₹69,089).
- Card, Wallet, PhonePe, GPay and COD are all in a similar ₹57k–₹69k band, i.e. UPI's dominance comes at the expense of every other mode roughly equally, not one specific competitor.

5.5 Trend / seasonality

- January (₹2,82,501, 756 orders) and February (₹2,70,809, 711 orders) are almost identical in both order count and revenue — there is no strong month-on-month growth or decline within the observed window.
- The steep drop shown for March on the dashboard (~₹10,682) is not a real seasonal decline — the extract only contains a single day of March data (1 Mar 2025), so the fall is a data-coverage artifact, not a business trend. This should be called out whenever the dashboard is presented, to avoid a false 'sales are dropping' conclusion.

5.6 Operational / fulfilment issues

- 16.3% of all orders end in Cancelled, Failed or Returned status rather than Delivered — a meaningful share of gross demand that never converts to realized revenue.
- Cancellation/failure rates are highest on NetBanking (22.0%) and Card (19.4%), and lowest on PhonePe (12.4%) — the payment experience itself may be contributing to failed orders on some modes.

- Product substitution happens on 48.5% of orders overall, and is highest for Essentials (59.4%), Veg (56.8%) and Baby (53.2%) — these categories are the most exposed to stockouts.
- Refunds total ₹72,763 across the period, concentrated in Stockout (47 cases), Delay (44) and Customer Cancel (40) as the leading reasons — inventory availability and delivery delay are the two biggest controllable drivers of lost revenue.

6. Business Recommendations

- Protect inventory for Essentials, Veg and Baby: these categories have the highest substitution rates (53-59%), meaning customers frequently don't get what they ordered. Tighter demand forecasting and safety stock at the warehouse level (especially Andheri, Saket and Koramangala, the busiest hubs) should reduce Stockout-driven refunds.
- Double down on Beverages, Frozen and Veg in marketing and shelf placement, since they are already the top revenue categories — small share gains here move the total more than equivalent effort on Bakery or Baby.
- Reduce dependence on Bulk customers by building a New/Returning customer growth motion (loyalty offers, first-order incentives): today Bulk accounts carry ~83% of revenue, which is a concentration risk if even a few large accounts churn.
- Investigate NetBanking and Card failure/cancellation rates (19-22%, the two highest of any payment mode) — a payment-gateway or checkout UX issue on these rails could be silently costing conversions.
- Fix the City capture gap so every order records a region; ~2% of orders currently can't be attributed to Bangalore, Delhi or Mumbai, which understates true regional performance.
- Treat the March figures as directional only until a full month of data is available, and refresh the trend chart monthly rather than drawing conclusions from partial periods.
- Standardize source data entry (payment mode spelling, category naming, price formatting) at the point of capture — a large share of the cleaning effort here was undoing avoidable formatting inconsistency, which recurs every reporting cycle unless fixed upstream.

7. Conclusion

Across categories, segments, regions and payment modes, Zepto's sales base over Jan–Mar 2025 is broad-based rather than concentrated in one dimension: revenue is fairly evenly split across the three cities and six customer segments, while Beverages, Frozen and Veg lead at the category level and UPI clearly leads on payment mode. The headline month-on-month trend is flat (January ≈ February), and the apparent March decline is a partial-data artifact rather than a genuine seasonal drop.

The more actionable findings are operational: roughly one in six orders fails to deliver, product substitution affects nearly half of all orders and is concentrated in a few categories, and refunds are driven predominantly by stockouts and delivery delays. Addressing inventory availability for the highest-substitution categories and understanding the elevated failure rates on NetBanking and Card payments offer the clearest near-term opportunities to convert existing demand into realized revenue, while the dashboard delivered alongside this document gives the business a repeatable way to monitor these cuts going forward.